

Renu Shekhawat

Stat-Math Unit, Indian Statistical Institute, 8th Mile, Mysore Road, RVCE Post, Bengaluru - 560 059
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Education

Indian Statistical Institute, Bangalore Centre
Philosophiae Doctorate in Mathematics
Supervisor : **Prof. Soumyashant Nayak**

Bengaluru, India
Jul 2019-present

University of Rajasthan
Master of Mathematics

Jaipur, India
Aug 2016 - Aug 2018

Sophia Girls' College
Bachelor of Mathematics

Ajmer, India
Jul 2013 - Jul 2016

Research Interests

Matrix analysis, Functional analysis, Operator algebras

Publications / Preprints

- S. Nayak, R. Shekhawat; *On the Convergence of the Normalized Power Sequence of Spectral Operators on Hilbert Space* (2024), available at : <https://arxiv.org/abs/2410.16318>
(Accepted for publication in JOT)
- S. Nayak, R. Shekhawat; *On the Jordan-Chevalley-Dunford decomposition of operators in type I Murray-von Neumann algebras* (2025), available at : <https://arxiv.org/abs/2506.17227>
(Submitted for publication).

Thesis

On the Jordan-Chevalley-Dunford Decomposition of Certain Classes of Operators and Convergence of Their Normalized Power Sequences. (Submitted in June 2025).

Fellowships / Scholarships

- **Junior Research Fellowship**

(Declined due to acceptance of the Institute fellowship at ISI)

- University Grants Commission, Government of India. Dec 2018
- Council of Scientific and Industrial Research, Department of Science and Technology, Government of India. Jun 2017

- **Inspire Scholarship for Higher Education**

Department of Science and Technology, Government of India. Jul 2013 - Jul 2018

Teaching Assistant / Grader

• Real Analysis, B.math(I) - Semester I Instructor : Prof. B. V. Rajarama Bhat	Aug 2025 - Nov 2025
• Function spaces, B.math(III) - Semester I Instructor : Prof. Soumyashant Nayak	Aug 2023 - Nov 2023
• Complex Analysis, M.Math(I) - Semester II Instructor : Prof. Ramesh Sreekantan	Jan 2022 - May 2022
• Complex Analysis, B.Math(III) - Semester I Instructor : Prof. Mathew Joseph	Sep 2021 - Dec 2021

Conferences / Workshops Attended

• School on Invariant Structures in Group Actions Research and Conference Centre of IMPAN, Będlewo	Będlewo, Poland Jan 11 - 16, 2026
• Young Mathematicians in C^* Algebras (YMC*A) 2025 University of Southern Denmark (SDU), Odense	Odense, Denmark Jul 21 - 25, 2025
• International Workshop on Operator Theory and its Applications (IWOTA) 2025 University of Twente, Enschede	Enschede, Netherlands Jul 14 - 18, 2025
• 20th Workshop: Noncommutative Probability, Operator Algebras and Related Topics, with Applications Research and Conference Centre of the Institute of Mathematics of the Polish Academy of Sciences, Będlewo	Będlewo, Poland Jul 6 - 12, 2025
• Young Mathematicians in Operator Algebras Indian Statistical Institute, Delhi Centre	Delhi, India Mar 24 - 28, 2025
• Conference on Operator Algebra and Related Topics (COART) Indian Institute of Technology, Bombay	Mumbai, India Feb 24 - 29, 2025
• Noncommutative Mathematics and Applications (NCMA) Indian Statistical Institute, Bangalore Centre	Bengaluru, India Oct 24 - 26, 2024
• Young Mathematicians in Operator Algebras Indian Statistical Institute, Delhi Centre	Delhi, India Feb 26 - Mar 02, 2024
• International Conference on Spectral and Approximation Theory (ICSAT) Cochin University of Science and Technology, Kerala	Kochi, India Nov 27 - 30, 2023
• Conference on Functional Analysis and Related Topics (CFART) Indian Institute of Technology, Bombay	Mumbai, India Feb 21 - 25, 2023
• IWM (Indian Women and Mathematics) Annual Conference Indian Institute of Science Education and Research, Pune	Pune, India Dec 27 - 29, 2022

- **Indo-French Centre for Applied Mathematics (IFCAM)**

Summer School on Mathematical Aspects of Quantum Mechanics
Indian Institute of Science, Bengaluru

Bengaluru, India
Jun 01 – 12, 2022

Talks

- **Invited:**

- As an *early Career Researcher* in **Simons Semester**

“Invariant Structures in Groups, Spaces, and Algebras”

Institute of Mathematics of the Polish Academy of Sciences, Warsaw

Jan 11 - Feb 16, 2026

- For a talk, *A spatial form of Yamamoto’s theorem for spectral operators*,
Analysis and Probability Research Group (APRG) Seminar,
Indian Institute of Science, Bengaluru

Sep 22, 2025

- **Contributed Talks :**

- *A stronger form of Yamamoto’s theorem on singular values*

IWOTA 2025, University of Twente, Enschede

Jul 18, 2025

- *Jordan-Chevalley-Dunford decomposition in Type I Murray-von Neumann algebras*

20th Workshop: Noncommutative Probability, Operator Algebras and Related Topics,
with Applications, Będlewo

Jul 10, 2025

- *Convergence of the normalized power sequence for spectral operators*

COART, Indian Institute of Technology, Bombay

Feb 27, 2025

- *On Matrices over $C(X)$ for X a Stonean space*

ICSAT, Cochin University of Science and Technology, Kerala

Nov 27, 2023

- **Seminar Talks :**

- *An elementary computational approach to the Jordan-Chevalley decomposition and the Jordan canonical form.*

PDF - RS Annual Symposium, Indian Statistical Institute, Bangalore Centre

Jan 31, 2025

- *Convergence of the normalized power sequence for spectral operators*

QP - Day at ISI, Indian Statistical Institute, Bangalore Centre

Jan 15, 2025